WILLIAM D. PERREAULT, JR. and LAURENCE E. LEIGH*

Most research related to the reliability and validity of marketing measures has focused on multi-item quantitative scales. In contrast, little attention has been given to the quality of nominal scale data developed from qualitative judgments. Judgment-based ("coded") nominal scale data are important and frequently used in marketing research—for example, in analysis of consumer responses to open-ended survey questions, in cognitive response research, in meta-analysis, and in content analysis. The authors address opportunities and challenges involved in evaluating and improving the quality of judgment-based nominal scale data, with specific emphasis on the use of multiple judges. They review approaches commonly used in other disciplines, then develop a new index of reliability that is more appropriate for the type of interjudge data typically found in marketing studies. Data from a cognitive response experiment are used to illustrate the new index and compare it with other common measures. The authors conclude with suggestions on how to improve the design of studies that rely on judgment-coded data.

Reliability of Nominal Data Based on Qualitative Judgments

In the last 15 years, significant attention has been directed to the quality of measures used in marketing research. The shift to an emphasis on quality was perhaps most clearly delineated by publication of such methodological contributions as Peter's review articles on reliability (1979) and construct validity (1981) and Churchill's (1979) paradigm for better measure development in marketing, but attention to measurement quality has extended into empirical work as well. For examples of measurement work spanning different substantive areas of marketing, see the collection of papers edited by Peter and Ray (1984).

The Quality of Judgment-Based Qualitative Data

Despite much attention to the quality of quantitative composite scales, marketing researchers have given only sparse consideration to the quality of nominal scale data developed from qualitative judgments. Such data are developed when a researcher devises a set of exhaustive and mutually exclusive classifications for a variable of interest, then asks one or more judges to evaluate a set of observations (responses) and code each observation into the most appropriate classification. The lack of attention in this area is notable given that judgment-based nominal scale data are important and frequently used in both academic and applied marketing research—for example, in analysis of consumer responses to open-ended survey questions, in cognitive response research, in protocol analysis, in meta-analysis, and in content analysis. The integrity of conclusions derived from such approaches depends on the quality of the judgment-based coding of the data on which any subsequent analysis builds.

Purpose and Organization

For these reasons, the purpose of our article is to address ways of evaluating and improving the quality of judgment-based nominal scale data in both applied and scholarly marketing research. More specifically, our objectives are to (1) synthesize the advantages and limitations of major approaches commonly used in other disciplines to evaluate judgment-based nominal scale data, (2) develop a parsimonious new approach for estimating

*William D. Perreault, Jr. is the William R. Kenan, Jr. Professor of Business Administration, University of North Carolina–Chapel Hill. Laurence E. Leigh is Stephens Associate Professor of Marketing, Birmingham Southern College.

The authors appreciate helpful criticisms and suggestions provided by Gary Armstrong, Merrie Brucks, Morgan Jones, Cheri Marshall, Charlotte Mason, George Milne, Fritz Russ, and three anonymous *JMR* reviewers. The authors are grateful for research support provided by the Business Foundation of North Carolina through the UNC Graduate School of Business Administration.

Journal of Marketing Research
Vol. XXVI (May 1989), 135–48

reliability that is more relevant to the type of data used in marketing studies, and (3) suggest ways to improve the design of research that involves judgment-based nominal measures.

We first highlight some of the important areas where judgment-based nominal scale data are used in marketing. Then we briefly review methods that have been developed and used in other behavioral disciplines to assess the quality of such measures. We examine Cohen's kappa—the most widely used coefficient of interjudge reliability and the basis on which most other contemporary approaches have been developed—and explain why kappa and its variations are inappropriate in most marketing research applications. To overcome the major limitations of available approaches, we develop a new model to estimate reliability of judgment-based data. We illustrate the application of the index with data from a cognitive response experiment. We conclude with recommendations about the design and reporting of marketing studies that rely on judges to code data.

BACKGROUND

Measurement Quality and Nominal Data

Though marketers' attention to measurement problems with quantitative scales certainly has been justified, the relative lack of attention to measurement quality problems with data that are not amenable to the multi-item, quantitative scale approach is difficult to explain. As Perreault and Young (1980) note, many of the variables or constructs of interest to marketers simply do not "measure up" at the interval or ratio scale level and instead involve categorical data that logically can be measured only at the nominal level. Further, the publication of many articles on data analysis models for use with nominal scale data attests to an interest in the challenges of research that relies on such data (cf. Green, Carmone, and Wachspress 1977; Hoffman and Franke 1986; Perreault and Barksdale 1980).

Measurement quality is not likely to be a major problem with every type of nominal scale data. For example, most respondents to a survey can accurately check their current marital status or what brand of car they own. Measurement quality—and more generally objectivity—is more likely to be a problem in research that requires judgment-based coding of observations into classifications. This general approach is used in many areas of marketing research and some of the most common applications are highlighted next.

Use of Judgment-Based Data in Marketing

Survey research. The most common use of judgment-based nominal scale data, especially in *applied* marketing research, is in the analysis of consumer responses to open-ended survey questions. In some cases, coding categories are developed in pretests to isolate an appropriate set of fixed-response categories to include on a final questionnaire. In other cases, however, the responses of a very large sample must be coded—especially if the purpose of the research is to understand how consumers respond without the aided recall provided by a set of precoded response alternatives. In yet other survey situations, interviewers are asked to code certain information (such as the respondent's race) directly from observation, without asking the respondent.

Content analysis. Content analysis procedures often are used in marketing research. Researchers who do quantitative analysis based on content analysis of oral or printed communications usually must first develop nominal data based on classification systems and qualitative judgments (Holsti 1969). Content analysis approaches are used in a wide variety of substantive research areas, including advertising research (Armstrong 1984), corporate communications (Bettman and Weitz 1983), and analysis of personal selling interactions (Farley and Swinth 1967).

Consumer behavior research. Judgment-based coding is now common in cognitive response research, but Wright's (1973) work on the processes mediating acceptance of advertising was one of the pioneering efforts in this area. Similarly, the Bettman and Park (1980) coding scheme stimulated increased use of judgment coding in research to analyze decision-making protocols. Researchers are becoming increasingly sensitive to the fact that the coding process is central to theory tests in these research areas (Biehal and Chakravarti 1982; Franke and Didow 1987; Lammers 1985).

Meta-analysis. Creative applications of meta-analysis procedures are leading to new insights about bodies of research in a variety of substantive areas (see, e.g., Assmus, Farley, and Lehmann 1984; Churchill et al. 1985; Fern, Monroe, and Avila 1986; Peter and Churchill 1986). However, meta-analysis approaches can be effective only if the researcher has first developed judgment-based, nominal scale coding schemes that bring structure to the diverse research studies in an area.

Convergence Among Judgments

There are no easy ways to ensure the reliability, validity, or objectivity of judgment-based nominal data, as *judgment* is intrinsic to the process. However, despite the sparse attention to these challenges by marketing researchers, extensive work has been done in other behavioral disciplines as reflected by review and methodological articles in sociology (Crittenden and Hill 1971; Krippendorff 1970), biomedical research (Koran 1975; Landis and Koch 1977), counseling (Lawlis and Lu 1972), education (Brennan and Prediger 1981), clinical psychiatry (Dewey 1983; Helzer et al. 1977; Maxwell 1977), and other areas of psychology (Fleiss 1975; Kraemer 1979; Light 1971; Mitchell 1979). We do not attempt to cover all facets of this extensive literature here, but focus on the topics most central to marketing research and the new procedures we develop.

With judgment-based data, as with most other types of data, the general approach is to develop multiple mea-

sures of the same construct and then to evaluate how well the measures converge (Light 1971). One procedure is to have a judge code a set of observations and later recode them (perhaps after they have been intermixed with new observations). This procedure can be used to check the consistency of the coder's judgments, but it has the obvious limitation of not identifying bias attributable to the (perhaps consistent) idiosyncracies of the judge doing the coding (Mitchell 1979). As a result, with judgment-based coding schemes and ambiguous stimuli, the best approaches for improving the quality of data rely on evaluation of judgments of two (or more) independent coders. Most of the work in this area—regardless of the specific behavioral discipline—addresses evaluation of interjudge reliability.

Clearly the quality of judgment-based data—and interjudge reliability—depends on the skill and motivation of the judge(s). From the outset, however, it also depends on the adequacy of the underlying classification scheme, the operational definitions for coding categories, and the directions that guide the coding process. Though judges are unlikely to agree consistently (by chance) in coding a large number of responses if some problem is present, agreement among judges on how to code any individual observation does not ensure that the observation is coded correctly. On a given observation, two judges might respond with equivalent guesses to imprecise directions or an ambiguous classification scheme. Moreover, disagreements between judges on how to code some observations forces attention to the underlying problem: because the data are only at the nominal scale level, one cannot simply "average" different codes (with errors of measurement) from different judges to arrive at a composite measure for use in subsequent analysis. Thus, even if reliability is in general high, any lack of reliability—as reflected in disagreement among judges—poses a problem.

As this discussion implies, the utility of an estimate of interjudge reliability is not necessarily limited to *ex post facto* evaluation of coded data. Rather, researchers often need a diagnostic application (in pretests or on a subset of data early in the coding process) to determine whether the classification scheme, definitions, directions, and training (i.e., the coders) can be improved. Once a sufficient level of reliability is achieved, attention can be given to general implementation of the coding process (which may or may not involve multiple judges to code every observation).

Regardless of whether a preliminary estimate of interjudge reliability is to be used for diagnostic purposes to improve the research design or, alternatively, as a summary index to reflect the quality of the final coded data, a first step is to estimate interjudge reliability of data arising from the coding process. We address that step in the next section. After examining the estimation issues, we return to a discussion of how reliability information can be used to improve the quality of the overall coding process.

MEASURES OF INTERJUDGE RELIABILITY

The Interjudge Contingency Table

As a starting point, it is useful to consider the nature of the cornerstone data on which all of the different methods of estimating interjudge reliability are based. First, each judge codes (classifies) each observation into one of the k mutually exclusive categories for the variable of interest. Thus, over a sample of N observations, each judge's classifications give rise to a frequency distribution (i.e., the number of observations allocated to each of k categories). Two (or more) judges' codes then can be cross-tabulated, resulting in an interjudge contingency table such as the one in Figure 1. The coding categories for each judge are aligned so that diagonal cells in the matrix represent judgments on which both judges agree. The frequencies in the off-diagonal cells not only indicate the level of disagreement, but also help to pinpoint the nature of the disagreement.

Percentage of Agreement

In the marketing research literature, there is no accepted standard for evaluating or reporting the reliability of coded data—even if multiple judges are used. When researchers use multiple judges and evaluate the convergence of their coding, the most commonly used measure of interjudge reliability is the simple percentage of agreement between two (or more) judges (cf. Bettman and Park 1980). This measure is easy to calculate and its interpretation is intuitive. Unfortunately, however, its simple intuitive interpretation has the potential to be misleading. As Cohen (1960) noted long ago, simple percentage agreement statistics are likely to be influenced heavily by the number of coding categories (i.e., cells in the interjudge contingency matrix). The smaller the number of categories—and the more highly skewed the distribution of judgments between them (Mitchell 1979)—the greater the likelihood of higher agreement due to chance alone. This characteristic makes it difficult to set quality standards based on percentage agreement statistics and confounds comparisons across different variables.

Cohen's Kappa Is the Most Widely Used Measure

Cohen (1960, 1968) proposed his kappa index of interjudge reliability to overcome the shortcomings of the simple proportion of agreement statistics. Though kappa also has limitations, it is very important because it is the most widely used measure of interjudge reliability across the behavioral science literature. It is also the basis on which most other approaches and extensions have been developed.

Kappa explicitly recognizes the likelihood of chance agreement between judges and removes it from consideration. Cohen (1960) defined kappa (K) as

$$K = (F_o - F_c)/(N - F_c) \tag{1}$$

where:

Figure 1
THE INTERJUDGE CONTINGENCY TABLE

	Coded category	Judge 1					
		1	2	3	. . .	k	
Judge 2	1	Agree					$f_{2,1}$
	2		Agree				$f_{2,2}$
	3	Disagree[a]		Agree			$f_{2,3}$
	⋮				Agree		⋮
	k					Agree	$f_{2,k}$
		$f_{1,1}$	$f_{1,2}$	$f_{1,3}$	. . .	$f_{1,k}$	Marginal distributions

[a]Any off-diagonal cell represents a disagreement among judges; for example, observations in this cell are ones that were coded into classification 3 by judge 2 but into classification 1 by judge 1.

N = the total number of judgments made by each judge,
F_o = the number of judgments on which the judges agree, and
F_c = the number of judgments for which agreement is expected by chance.

Prior to Cohen's work, other measures of interjudge reliability had used this same general form (see, e.g., Goodman and Kruskal 1954). The basic difference in Cohen's measure is his approach for estimating the F_c (chance agreement) component. He argued that chance agreement is a function of the marginal probabilities—the distribution of responses across categories by each judge. Specifically, drawing on the multiplicative law of probability, he estimated the probability of "chance agreement" between two independent judges as the product of the (independent) marginal probabilities.

Though subsequently it was shown that kappa can be viewed in terms of a population model of reliability (Kraemer 1979), Cohen simply defined kappa in terms of the procedures used to compute the statistic. Basically, he saw kappa as a "well-behaved" index. An index value for kappa of +1.0 indicates perfect agreement between judges on all cases and a value of 0.0 indicates no agreement other than that expected by chance. Negative values indicate greater than chance levels of disagreement between the judges. Negative values of kappa are not bounded by −1.0, but by itself this fact is not a practical drawback to interpretation because *disagreement beyond chance* is unlikely to arise in actual research; if it does, precise calibration of reliability is premature.[1]

Cohen (1960, p. 44) also proposed an estimate of the confidence interval for assessing the statistical significance of estimated levels of kappa against the null hypothesis of "no agreement other than by chance between the judges." Later this estimate was recognized to be overly conservative and inexact because it was based on the contradictory assumptions of fixed marginal totals and binomial variation of cell frequencies. However, refined computational procedures ultimately were proposed that avoided these problems (Everitt 1968; Fleiss, Cohen, and Everitt 1969). These revisions addressed the technical imprecision of Cohen's original proposal, but one should keep in mind the basic limitation of the hypothesis being tested. As Cohen (1960, p. 44) himself noted,

> . . . it is generally of little value to test K for significance . . . to know merely that K is beyond chance is trivial since one usually expects much more than this in the way of reliability in psychological measurement. It may, however, serve as a minimum demand in some applications.

Though the substantive hypothesis tested by constructing a confidence interval about kappa is a limitation, other basic properties of the kappa statistic are more problematic.

Limitations of Cohen's Kappa for Marketing Applications

In spite of the popularity and frequent application of kappa in many disciplines, it has serious and well documented drawbacks that make it inappropriate for many research problems—including most marketing research applications. Brennan and Prediger (1981) provide a useful technical review of the problems and limitations of kappa; their article is recommended reading. Basically, the limitations stem from the assumptions underlying Cohen's reliance on the multiplicative law of probability in estimating the chance agreement component of kappa. As we discuss in more detail shortly, the result is an overly conservative measure of reliability, which in turn makes it difficult to compare values of kappa with values from other types of reliability indices, such as Cronbach's alpha.

Kappa is conservative because of the way it estimates

[1]Sometimes kappa is not as "well-behaved" as would be desirable. As Brennan and Prediger (1981) discuss in more detail, with asymmetric marginal probabilities the maximum value of kappa can be less than 1.0.

Table 1
SAMPLE INTERJUDGE CONTINGENCY DATA WITH NONUNIFORM MARGINAL DISTRIBUTIONS (N = 100)

	Coded category	Judge 1		
		1	2	
Judge 2	1	81	9	.9
	2	9	1	.1
		0.9	0.1	Marginal probabilities

the agreement cell frequencies to be expected by chance. *Specifically, it takes as given the marginal distributions between categories exhibited by each of the judges.* A simple example illustrates the impact of this approach. For the problem in Table 1, each of the judges places 90% of the observations in the first category and 10% in the second. On the basis of this information, we would calculate kappa's chance agreement cell frequencies as $F_c = 100[(.9)(.9) + (.1)(.1)] = 82$; Cohen's measure only "gives credit" to the judges for agreements that are beyond this chance level. Thus, because the observed frequency of agreement is also 82, the value of kappa in this case is 0. Note, however, that this view of chance *ignores* the fact that both judges agree that a much higher proportion of cases fit into the first category than into the second. Indeed, kappa is intended merely to measure the degree to which agreement about which cases to put into which category has been achieved beyond chance levels *for a given set of marginal probability distributions associated with each judge.*

Cohen recognized this limitation of his measure and for the applications he had in mind it was not likely to be serious. He was concerned mainly with psychological applications for which there often would be clearly established prior knowledge of the likely distribution of observations across cells. For example, in rating a sample of patients for the exhibition of symptoms of psychologically disturbed behavior, it is reasonable to assume that judges (e.g., clinical psychiatrists) will assign *a priori* a relatively small proportion of cases into a category known to be rare in the population.

In contrast, in most marketing research studies (and in many other areas of applied research), there is no *a priori* knowledge of the likely distribution of responses—nor a basis for assuming what the distribution between categories should be. In fact, in many situations, getting an estimate of the distribution of responses across categories is a key reason for doing the research in the first place. For example, the widely reported periodic studies of the Index of Consumer Confidence are based on frequency distributions developed from coding of open-ended responses (Didow, Perreault, and Williamson 1983). For a measure of interjudge reliability not to consider any agreement among judges on the marginal distributions is not only conservative, but also potentially misleading (Brennan and Prediger 1981).

Other Major Work on Interjudge Reliability

The limitations of kappa have prompted researchers to propose several extensions and alternative formulations of interjudge reliability. For example, Fleiss (1971) generalized kappa to the case in which a sample of observations (i.e., consumer responses, patients to be diagnosed, etc.) are judged and coded on a nominal scale by the same number of judges, but the judges rating one stimulus are not necessarily the same as those rating another.

At about the same time, Light (1971) proposed various extensions of Cohen's kappa to different types of interjudge evaluations; for example, he argues that his statistics can be applied to measure the *pattern* of agreement among several judges and to measure agreement of judges against an *a priori* standard. Light also reports the results of a Monté Carlo investigation of the properties of his generalized statistics to serve as a needed aid in interpretation.

Maxwell (1977) proposed an alternative measure of interjudge reliability for binary classifications in clinical psychiatry and Janes (1979) subsequently extended Maxwell's index for multilevel classification problems. Their logic is interesting because it takes into consideration the likelihood that some agreement and some disagreement occurs by chance. However, the index they propose does not have the invariant and easily interpretable bounds of most reliability statistics. Perhaps that is one reason it has not been widely adopted by others.

Landis and Koch (1977) proposed a model of interobserver agreement based on the weighted least squares estimation of the relationships between categorical variables. The complexity of implementing the computations required for their approach probably has slowed diffusion of their work, but the large sample sizes required and other assumptions of weighted least squares analysis also limit the applicability of their approach in many research situations.

Brennan and Prediger (1981) reviewed some reliability indices that directly address Cohen's assumption of fixed marginal distributions. For example, they discussed an index originally proposed by Bennett, Alpert, and Goldstein (1954). Brennan and Prediger designated the index as K_n because it is computed by the same formula as Cohen's kappa (equation 1), except that the chance agreement component is estimated as N/k, where k is the number of categories for a variable being coded. This adjustment is based on the logic that when the judges do not know the true marginal distributions in advance, they are equally likely to code an observation into any individual category. Long ago, Scott (1955) pointed out that this index is based on the assumption that all categories have equal probability of use by both coders–and he argued that such an assumption is unwarranted for most types of behavioral and attitudinal research. Brennan and Prediger acknowledged that criticism, but adopted the point of view that the index is useful in some situ-

ations because $1/k$ is a "useful lower limit for the unknown probability of agreeing by chance" (Lawlis and Lu 1972). Unfortunately, however, they did not propose a way to determine how far from correct the lower limit might be. In addition, to the concerns raised in that history-spanning debate, confidence intervals for K_n appear to be subject to the same problems as confidence intervals for kappa; that may be why Brennan and Prediger did not propose any type of confidence interval interpretation for K_n.

Each of these research efforts has added new dimensions to understanding issues of interjudge reliability and each has addressed at least some of the limitations of kappa. However, Cohen's kappa has remained as the most commonly used measure of interjudge reliability. In an article on coefficients of agreement, Dewey (1983) argues that Cohen's kappa should still be "the measure of choice." Yet, given the problems with kappa (and the limitations and interpretation difficulties of other major approaches for evaluating interjudge reliability), it makes sense to consider an estimate of interjudge reliability (and associated confidence intervals) that is more appropriate to the type of data typically encountered in marketing contexts).

AN ALTERNATIVE MEASURE OF RELIABILITY

Our approach for estimating the reliability of interjudge data is very different from the approach used by Cohen and approaches used in most other research in this area. One major difference is that we do not contrast observed agreement between judges with some estimate of chance agreement, but rather develop an explicit model of the level of agreement that might be expected given a true (population) level of reliability. This approach makes it possible to compute directly a sample-based estimate of reliability and the associated standard error of the estimate. The index has a direct, intuitive interpretation and results in values between 0.0 and 1.0—like most reliability measures used with quantitative data. Further, the assumptions involved in arriving at an estimate are reasonable, explicitly set forth in developing the model, and the index can be shown to be robust to even extreme violations of the assumptions.

As a starting point to understanding the new approach, it is useful to designate a term, F_e, that is the *frequency of agreement expected due to the true (population) reliability* of the overall coding process. Conceptually, reliability here can be thought of as the percentage of the total responses (observations) that a typical judge could code consistently given the nature of the observations, the coding scheme, the category definitions, the directions, and the judge's ability. Note that this definition explicitly acknowledges the various factors that influence the reliability of the coding process (as discussed in the introductory sections of the article).

Agreement Expected Due to Reliability

In this section, we develop a model of the level of agreement that might be expected between a pair of judges given a true level of reliability of the coding process. Though the true level of reliability is unknown, we specify an index, I_r, that represents the reliability. Consistent with the conceptual definition provided before, the operational range of values for I_r (the reliability index) is between 0.0 (no reliability) and 1.0 (perfectly reliable). Thus, for a coding process with reliability I_r a given judge will reliably code NI_r observations and will allocate $N(1.0 - I_r)$ unreliable judgments as guesses. In this simple framework, the total number of judgments, N, made by a judge is simply the sum of the reliable judgments and the unreliable ones. Now, to understand better what we might expect in terms of agreement between judges given reliability of I_r, let us consider how the reliable and unreliable judgments will be allocated to different cells in an inter-rater contingency table (such as the one in Figure 1).

First, let us consider the reliable judgments. As noted before, each judge makes I_r percent of the judgments reliably. However, because each judge operates independently of the other, the specific judgments on which one judge is reliable are independent of the other judge's reliable judgments. We therefore would expect (estimate) the overall number of observations for which the two judges are both "reliable" and "agree" to be equal to

$$\text{(2)} \qquad \text{Number of reliable judgments on which judges agree} = NI_r^2.$$

Note that here we are not making an assumption about the marginal distributions of the judgments, but rather focusing on the notion that there is some true (but perhaps imperfect) level of reliability implicit in the independent judgments.

We have considered the NI_r^2 judgments on which both judges are reliable and agree. Now, let us turn to the remaining $N(1 - I_r^2)$ judgments. On some of these judgments, both judges are reliable, but they do not agree. Those reliable-disagree judgments would be in cells off the agreement diagonal. What remains are classifications on which either or both judges are unreliable. Some of these unreliable judgments may result in agreement and others may not, but on how many of the unreliable guesses should the judges' classifications agree?

To answer this question, recall that if each judge can code an observation in k different ways, there will be k^2 possible combinations of judgments. Of that total, k of the combinations would result in agreements (i.e., combinations that are on the agreement diagonal). Consistent with the fact these judgments are unreliable, we should expect that (within sampling variance) the ratio of "guesses" on which the judges agree to the total number of judgments will be proportionate to the ratio of possible ways in which they could agree to the possible number of combinations, which is k/k^2. Thus, unreliable judgments for which there is agreement should be in a proportion of k/k^2 to the total remaining $N(1 - I_r^2)$ judgments. Specifically, the number of unreliable agreements should be

$$N(1 - I_r^2)(k/k^2) = N(1 - I_r^2)/k. \quad (3)$$

Without making any assumptions about the marginal distributions of the judgments in the interjudge contingency table, we have shown how we might expect the reliable and the unreliable judgments to be allocated. We now can combine these two separate expectations to arrive at an expectation about the total number of judgments in the k cells of the agreement diagonal of the contingency table. We simply add the expectations for "reliable" allocations (equation 2) and "unreliable" allocations (equation 3), which results in

$$F_e = \{NI_r^2\} + \{N(1 - I_r^2)/k\} \quad (4)$$

or, after rearranging terms,

$$F_e = N(I_r^2(1 - 1/k) + 1/k). \quad (5)$$

Thus, we have arrived at an estimate of the expected frequency of agreement given a true level of reliability for the coding process.

Estimating Reliability

Having derived the expected frequency of agreement, we can estimate the value of I_r, our index of reliability. Specifically, given the true level of reliability, the observed frequency of agreement between judges, F_o, should be equal (within sampling variance) to the expected frequency of agreement, F_e. Hence we can set F_e equal to F_o in equation 5, substitute terms, and solve for I_r^2.

$$F_o = N\{[I_r^2(1 - (1/k))] + (1/k)\}, \quad (6)$$

$$(F_o/N) = \{I_r^2(1 - (1/k))\} + (1/k),$$

$$(F_o/N) - (1/k) = I_r^2[1 - (1/k)],$$

$$[(F_o/N) - (1/k)]/[1 - (1/k)] = I_r^2$$

$$[(F_o/N) - (1/k)][k/(k - 1)] = I_r^2.$$

Thus,

$$I_r = \{[(F_o/N) - (1/k)][k/(k - 1)]\}^{.5}, \quad \text{for } F_o/N \geq 1/k. \quad (7)$$

Note that there is no real number solution to equation 7 if the observed proportion of agreement (F_o/N) is less than $1/k$. In that case, however, there is no need to estimate reliability, because reliability is at the value 0. That condition implies that the percentage agreement is less than would have been expected with both judges making totally random assignments of observations to the various categories. Therefore, we can supplement equation 7 with

$$I_r = 0, \quad \text{for } F_o/N < 1/k. \quad (8)$$

Though it is simple to estimate reliability for empirical data by substituting values in equation 7, Figure 2 provides a broader overview of the behavior of the reliability index across different levels of observed agreement for different numbers (ranging from two to nine) of coding categories. Use of the plot to estimate reliability from empirical data is straightforward. For example, with 80% agreement between judges and three coding categories for a variable, the estimated reliability is .84.

Figure 2 shows that for a given level of observed agreement, the estimate of reliability becomes higher as the number of possible response categories increases. This general relationship should be no surprise. As discussed before (equation 3), the number of possible response categories influences the number of guesses on which we expect the judges to agree. However, Figure 2 makes the nature of this relationship more specific: as the number of coding categories increases, reliability increases at a decreasing rate. Beyond nine categories, the curves tend to converge—so little additional information would be provided by plotting larger number of categories.

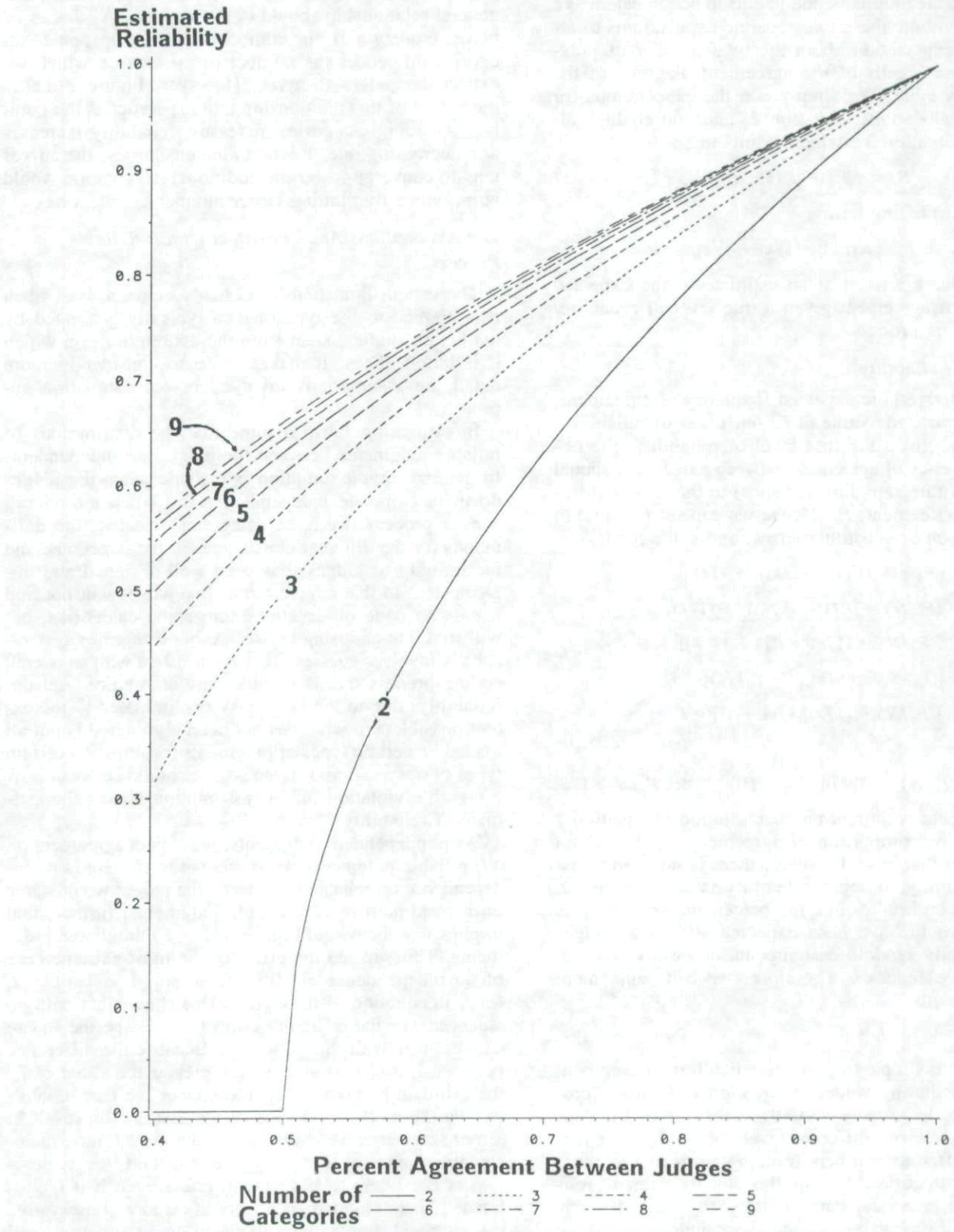
The Assumptions Are Consistent with a Reliable Process

The various limitations of Cohen's kappa, at least when it is applied to the type of data typically generated by marketing studies, stem from the assumptions on which it is based. Thus, it makes sense to consider in more detail the assumptions involved in our estimation approach.

In estimating I_r we assume that the distributions of reliable judgments between the judges are independent. In general, this assumption is realistic when the judges do in fact operate independently and when the overall coding process (i.e., the categories selected, the definitions for the different classifications, the directions, and the training of judges) has been well designed and implemented. In that circumstance, two judges will not find it easy to code observations into some categories, but will find that choosing between other categories systematically involves guesses. This is consistent with an overall coding process that is reliable; any ambiguity (and unreliability) due to the basic task is eliminated to the extent possible and what has not been eliminated is not all loaded on certain types of judgments (or, thus, on certain types of observations). Even so, we should consider how a possible violation of this assumption affects the estimate of reliability.

With independent judgments, we expect agreement on NI_r^2 reliable judgments. However, even if there is *no* independence on reliable judgments, the judges would agree on a maximum of NI_r reliable judgments; further, that implies that they would make $N(1 - I_r)$ unreliable judgments. Thus, to see the effect of the most extreme case of no independence at all, we can simply substitute I_r for I_r^2 in equations 4 through 6. This shows that with *no independence* the reliability estimate would be the square of what it is with independence. Because the difference between I_r^2 and I_r is smaller the greater the value of I_r, the estimate is more robust the greater the true reliability. Furthermore, violations of the assumption result in a more conservative estimate of reliability. Thus, in acceptable ranges of reliability, the independence assumption is not likely to have much effect even if it is violated. If it is violated, it results in a more conservative estimate and thus is more likely to direct attention (and

Figure 2
ESTIMATED RELIABILITY FOR DIFFERENT PERCENTAGE AGREEMENT LEVELS AND DIFFERENT NUMBERS OF CODING CATEGORIES (FROM 2 TO 9)



corrective action) to the problems that might have caused the assumption to be violated; coding should be designed so that judges are independent. In sum, this assumption is reasonable and its effect on the behavior of the estimate is logically consistent with the intended use of the index.

We also assume that the judges are equally reliable. This assumption is reasonable because, as Cohen (1960) has noted, in research involving judges there is typically no criterion for the "correctness" of judgments and the judges are deemed *a priori* equally competent to make judgments.

To develop an estimate of how many of the unreliable judgments might be allocated to the cells on the agreement diagonal, we have assumed that the percentage of unreliable judgments allocated to all of the cells on the agreement diagonal is equal to the percentage of the total cells that are on the agreement diagonal, or $1/k$. In considering this assumption, it is important to see that it does not depend on any assumption about the marginal distribution of judgments (as is the case with Cohen's kappa). Further, it does not depend on the assumption that all categories have an equal probability of use by both judges, as is the case in the variation of kappa suggested by Brennan and Prediger (1981). More specifically, it does not assume that the probability of responses being in a particular cell is equal (i.e., that the probability of responses being in one of the k^2 cells is $1/k^2$). On the contrary, the assumption is compatible with a condition that might arise if one or both judges use some coding categories more frequently. In such a case, the implication is that some off-diagonal cells have more responses and others fewer; similarly, some cells on the agreement diagonal have more responses than other on-diagonal cells. In other words, the "high" frequency cells and "low" frequency cells are simply expected to average out. Furthermore, the higher the reliability, the more robust the index is to a violation of this assumption, and the index moves in the appropriate direction with violations of the assumption. To the extent that an unexpected (disproportionate) number of the guesses result in agreement (i.e., the ratio of agreeing unreliable judgments to total unreliable judgments is greater than $1/k$), the estimated index will be high. To the extent that the ratio is less than $1/k$ (i.e., a disproportionate number of guesses result in disagreements), the estimate will be low.

In summary, the assumptions underlying the index of reliability developed here are both straightforward and reasonable—and they are consistent with the concept of a more reliable process resulting in a higher index. Like the simple percentage agreement statistic, the values of the index range from 0.0 to 1.0, but the index accounts for differences that are due to different numbers of categories for different variables. Furthermore, interpretation is not confounded by a sensitivity to the marginal distributions (which makes kappa overly conservative in many applications). The conceptualization of the index is useful because it focuses on the reliability of the whole coding process—not just on agreement between judges *per se*. If the assumptions are seriously violated, there is likely to be a problem with some element of the coding process (directions, categories, etc.), even if the observed (percentage) agreement is relatively high. If these problems are severe, the index will be lower, signaling that corrective actions should be taken.

Confidence Intervals for Reliability Estimates

We have noted that the index I_r is an estimate, within sampling variance, of the true proportion of reliably coded observations. When one is sampling without replacement from an infinite (or, as a practical matter, very large) population, estimated proportions follow the binomial sampling distribution. Thus, by drawing on the normal approximation to the binomial distribution, we can estimate an approximate confidence interval for the reliability estimate. Specifically, if z_c is the critical value for the c percent confidence interval and N is the sample size, the upper and lower confidence limits for reliability are

$$\text{Limits} = (I_r + z_c^2/2N) \pm z_c[(I_r(1 - I_r)/N) + (z_c^2/4N^2)]^{.5}/(1 + z_c/N). \tag{9}$$

Conveniently, for large samples (e.g., $N > 30$), equation 9 reduces to

$$\text{Limits} = I_r \pm z_c[I_r(1 - I_r)/N]^{.5}. \tag{10}$$

More detailed discussions of the derivation of the confidence interval estimate given in equation 9—and how it relates to the familiar normal approximation of confidence intervals for binomial estimates given in equation 10—can be found in standard statistics texts (see, e.g., Spiegel 1961, p. 162–3).[2]

The normal approximation (equation 10) is better with larger sample sizes and when the reliability estimate is not close to 1.00. In practice, the approximation should be adequate when $I_rN > 5$ (Spiegel 1961, p. 124). When I_r is close to 1.00 (say, $I_r > .95$), there will be less concern about the precise size of the confidence interval and in that case one might simply use the lower limit of the interval around a .95 estimate as a conservative approximation.

As an example of the confidence interval approach, consider a research situation in which the reliability is estimated to be .87 based on a sample of 200 observations. If the researcher is interested in the 95% confidence interval, the critical value (z_c) would be 1.96 and thus from equation 10 the confidence limits would be

[2] A *JMR* reviewer suggested an alternative approach for estimating confidence intervals. Specifically, if the reliability index is conceptualized as a correlation, Fisher's r to z transformation could be used to obtain confidence intervals. First, the estimated index would be transformed to a z score and then confidence limits on the z value would be estimated. Next, the limits could be transformed back into the corresponding values on the r scale. A potential advantage of this approach is that the upper limit could not exceed 1.0. A potential disadvantage is that the model used to develop the index is based on a different conceptualization of reliability.

$.87 \pm 1.96(.87)(.13)/200 = .87 \pm .025$. Hence, in this situation the lower limit of reliability (at the 95% confidence level) would be .855.

Though it is simple to compute the deviation about the reliability estimate (equation 10) to get the upper and lower confidence limits, Figure 3 provides a broader overview of how that deviation varies for different levels of reliability and for different sample sizes. This graph is especially useful in planning the sample size required to ensure that the reliability of a set of coded measures exceeds some critical value, for example .80. It is also a useful reminder that reliability estimates that build on percentage agreement data are more precise with larger samples (Kraemer 1979). This point holds regardless of what index of reliability is used—and regardless of the availability of a confidence interval for an index. In other words, ignoring the issue of sampling variance does not make it go away.

Figure 3 does not show directly how the number of coding categories affects the confidence interval. However, that effect warrants consideration. As is shown in Figure 2, for a given level of observed agreement between judges, the reliability estimate increases as the number of coding categories increases. Further, from Figure 3 we can see that the standard error of the estimate decreases as the reliability estimate increases. Thus, for a given level of agreement, the standard error of the estimate of reliability will decrease as the number of coding categories increases. The magnitude of this effect, however, decreases as the observed level of agreement and the number of coding categories increase.

AN EMPIRICAL EXAMPLE

To illustrate the use of the proposed measure in an actual example and to show how it relates to others often used, we reanalyze data from an experimental study of children's responses to advertising messages (Brucks, Armstrong, and Goldberg 1988).[3] The Brucks, Armstrong, and Goldberg article details the purpose for and design of the study. Basically, two judges evaluated each "thought" expressed by each of the subjects on each of five variables ("dimensions"): relevance, form, target, origin, and polarity. There were three possible (nominal) response categories for the form and relevance dimensions. The target dimension involved eight possible categories and the origin dimension involved four categories. The polarity dimension was judged as positive, neutral, or negative. The categories for the polarity dimension might be viewed as being on an ordinal scale. However, here we simply treat this trichotomy as a nominal scale.

Table 2 summarizes the proportion of agreement between the judges for each dimension and gives alternative estimates of reliability—Cohen's kappa, Brennan and Prediger's K_n, our I_r, and the 95% lower confidence limit for I_r. From the table, we see that agreement between the judges is relatively high on each of the dimensions. Furthermore, as might be expected given the preceding discussion of kappa, the value of kappa is consistently lower than the percentage agreement statistic. However, it is also useful to recognize that the interjudge agreement on the different dimensions, as suggested by the rank order of these two measures, is very different. For example, the dimensions in the table are in descending order of percentage agreement and the relevance dimension is second with percentage agreement equal to .95. In contrast, the value of Cohen's kappa for the relevance dimension is the lowest of the five dimensions, .64. Part of this inconsistency is due to the fact that lower values of kappa occur when the marginal distributions for the judgment categories are more skewed.

The results in Table 2 also show that the values of kappa for different sets of judgments may be very similar in spite of large differences in the percentage agreement observed in the sample. For example, compare the statistics for the form dimension and the polarity dimension. The percentage agreement on the form judgment is very high (.98), but for the polarity judgments it slips to .85. Even so, the corresponding values of Cohen's kappa (.762 vs. .757) suggest no meaningful difference in interjudge reliability.

Table 2 also gives the probability level for the hypothesis test based on (the revision of) Cohen's proposed confidence interval. Keep in mind that the hypothesis is that there is no agreement except by chance. As might be expected from simple inspection of the percentage agreement, this hypothesis is overwhelmingly rejected in each instance.

Now consider the right side of Table 2, which gives statistics relevant to I_r and K_n. As might be expected given that the I_r index is not dependent on the judge's marginal distributions, the ordering of the different dimensions on the basis of this measure more closely parallels the ordering for the agreement proportions than does Cohen's kappa. Unlike the proportion of agreement measure, however, I_r takes into consideration the number of alternative coding categories (k) for a variable. Unlike Brennan and Prediger's K_n, it does not assume that all categories are equally likely to be used by each judge. Because of this assumption, K_n is likely to result in a deflated estimate of reliability, especially if the distribution of observations is not uniform across categories. The relative sizes of K_n *and* I_r in Table 2 illustrate this tendency.

Table 2 also shows the lower limit of the 95% confidence interval about the I_r reliability estimate; a confidence interval test for K_n is not provided because one has not been proposed in the literature. Note that for these variables the I_r estimate and the lower limit are close to each other—because the estimates are based on large samples. Both the index of reliability and the confidence interval approach suggest that the coding process is re-

[3]The authors are grateful to Gary Armstrong, Merrie Brucks, and Marvin Goldberg for their willingness to provide their data for use in this comparative analysis. The form variable discussed here was developed by them as part of their study, but it was not used in the published version of their article.

Figure 3

STANDARD ERROR OF THE RELIABILITY ESTIMATE TIMES 1.96 FOR DIFFERENT LEVELS OF RELIABILITY AND DIFFERENT SAMPLE SIZES

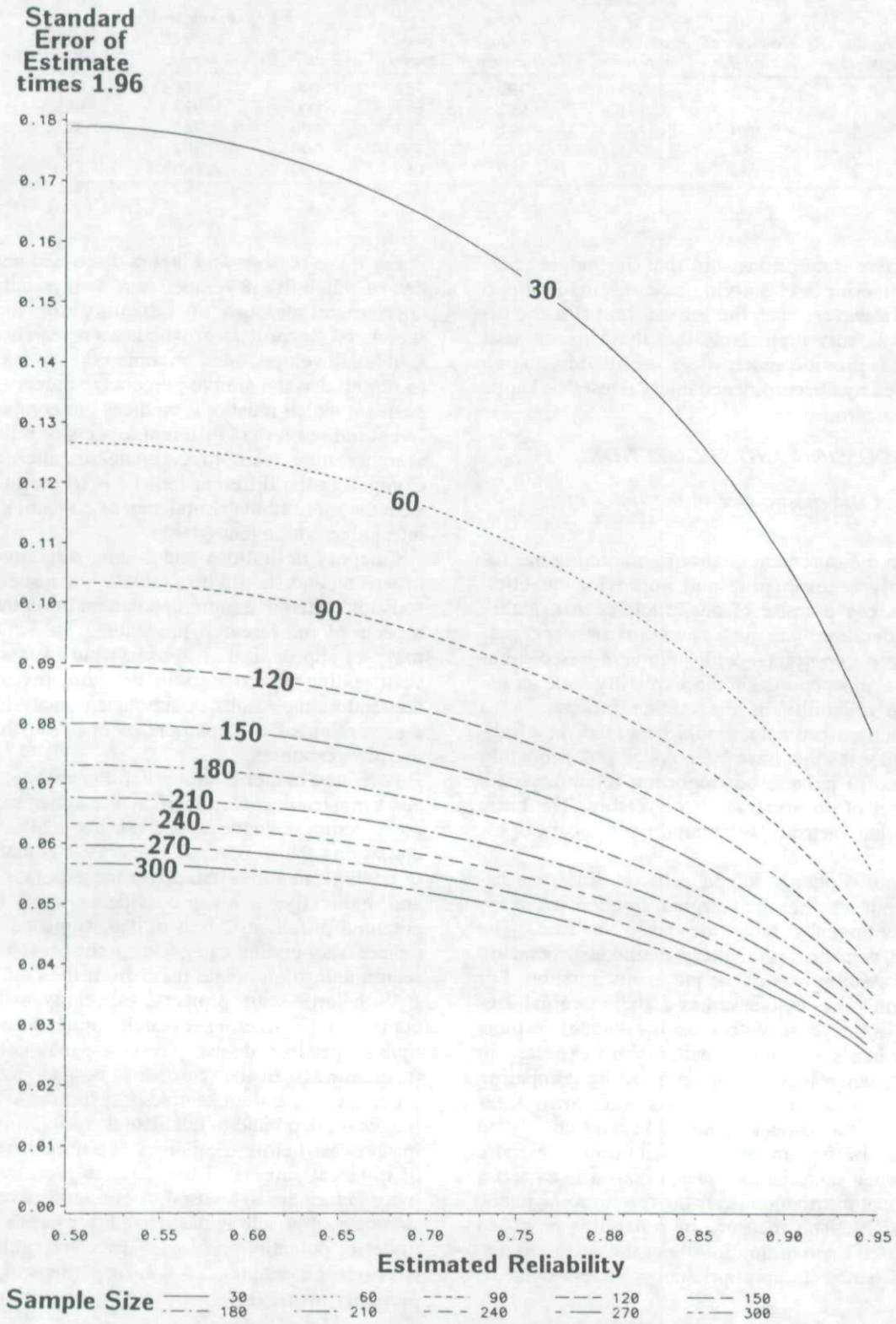


Table 2
ALTERNATIVE RELIABILITY ESTIMATES FOR CODED EVALUATIONS OF CHILDREN'S COGNITIVE PROCESSING OF ADVERTISEMENTS

Dimension	*Number of categories*	*Number of thoughts*	*Frequency of observed agreement*	*Percentage agreement*	*Cohen's kappa*	*Probability (kappa = 0)*	*Brennan & Prediger's kappa$_n$*	*I_r estimate of reliability*	*95% confidence lower limit of I_r*
Form	3	889	873	.982	.762	.000	.974	.986	.97
Relevance	3	982	934	.951	.643	.000	.927	.963	.95
Polarity	3	889	748	.841	.757	.000	.763	.873	.85
Target	8	888	703	.792	.733	.000	.762	.873	.85
Origin	4	889	667	.750	.656	.000	.666	.816	.79

liable for all five dimensions, but that the judges' performance is superior in classifying cases on the first two dimensions. However, even the lowest limit (for the origin variable) is very high. Note that these lower limit estimates for I_r provide much more useful information than is provided by the confidence interval test that kappa is greater than zero.

DISCUSSION AND SUGGESTIONS

Evaluation and Diagnostic Use of the Index of Reliability

Attention to measurement quality in marketing has focused primarily on evaluating and improving quantitative scales. A key premise of our article is that quality control considerations are just as important—and perhaps even more important—with judgment-based nominal data. Research reports should explicitly state an estimate of the reliability of the coding process. At a minimum, such an estimate should be based on a subsample of responses that have been coded independently by two (or more) judges, but reporting reliability data for the total set of observations is preferable. The basic question is what measure of reliability is most appropriate.

Variations of Cohen's kappa rely on different assumptions about whether the marginal frequencies in the interjudge contingency table are fixed or free. The strengths and weaknesses of these methods depend on how well the assumptions fit the judgment situation. For example, a measure that assumes fixed marginal frequencies (as is the case with Cohen's kappa) is most appropriate when set response patterns are expected in advance (i.e., when judges are evaluated by comparing their judgments against an established standard). Reliability estimates based on variants of kappa that rely on an assumption of free marginal distributions are more appropriate when there is no *a priori* reason to expect a specific marginal distribution (as is the case in some public polling research). The measure of reliability proposed here is not based on assumptions about the marginal distribution and hence its appropriateness is less context-specific.

For these reasons and others discussed before, the index of reliability developed here will usually be a more appropriate measure of reliability for marketing researchers. Regardless of whether a researcher reports the I_r index developed here or some other index, it is useful to report also the simple percentage agreement. It is the basis on which most other indices are computed. As different indices reflect different aspects of reliability, a researcher may want to compute an alternative index. Computing the different indices is trivial in comparison with the more fundamental step of collecting data for the interjudge contingency table.

Category definitions and coding directions should be in writing and should be available for inspection by persons interested in a more detailed audit on the qualitative aspects of the research procedure. This documentation may not appear to be necessary, but in many situations such coding directions will be more revealing for understanding the results of subsequent analysis than seeing the wording of the open-ended question that prompted the raw responses.

A related matter is that reliability estimates may present a misleading picture if a researcher retains one or more coding categories that are not really necessary in classifying the responses. The reason is that the number of coding categories can affect the estimate of reliability and, indirectly, its lower confidence limit. To avoid this potential pitfall, it is best to detect unused or otherwise unnecessary coding categories in the pretest stages of research and to eliminate them from the coding process.

With large-scale projects, especially in large-sample commercial marketing research applications, using multiple judges to code every response may not be practical (economical). In that case, it is best to start the coding process on a random sample of responses using multiple judges (coders) and to evaluate the reliability of the coding process before the judges begin independent coding of different subsets of the data. In fact, even if two or more judges are to be used in evaluating every response, starting with a subsample reliability analysis makes sense to detect potential problems. If the interjudge reliability is low (for example, $I_r < .8$, or perhaps .7 in more exploratory work) for a subsample of responses initially

coded and evaluated, corrective adjustments can be made early in the research process. For example, the researcher may need to clarify or redefine the operational rules that the judges use to code the data. Sometimes it is necessary to introduce new categories to accommodate responses that do not fit logically into the *a priori* classification scheme (and which therefore cause confusion among the judges and disagreements in coding). Further, evaluation of specific patterns of interjudge consistency (based on the off-diagonal cells of the interjudge contingency table) can help to identify sets of classifications that tend to be confounded or ambiguous. If the contingency table data are complex, simple inspection of the data may not be adequate for diagnostic purposes. However, in such situations the researcher might apply probabilistic latent class models (Dillon and Mulani 1984) or log linear models (Green, Carmone, and Wachspress 1977) to help detect subtle relationships (e.g., judge-category interactions).

When multiple coders have been used and disagreements on codes are identified, a decision must be made about what value to use for subsequent analysis. This decision involves a distinction between *quantitative* data from multiple judges and qualitative data from multiple judges. With quantitative data (assuming some acceptable level of inter-rater reliability), one typically develops a (weighted) composite of the different responses—basically viewing the individual judgments as indicators (with error) that are combined to produce a better single measure. That approach is not feasible with qualitative data. When three or more judges have been used, discrepancies can be resolved on a majority rule basis. In fact, this approach suggests a basis for resolving differences even when most responses are coded by only two judges. Specifically, for many types of judgment-based qualitative data, one can increase research objectivity while maintaining some economy by using three or more judges but dividing the total coding/judgment job so that there are initially only two judges for each response. Several simple methods can be used to randomize coding assignments across different pairs of judges. Then, after determining that the initial level of interjudge reliability is adequate, one can focus attention on the exceptions—the subset of responses that did not elicit identical judgments from the initial two independent judges. For those responses, a judge who was not previously involved can be used to provide another independent assessment. This approach is more labor intensive than just dividing up the coding task, but it is worth considering—especially if the sample of response is not very large, collecting the data is expensive (in relation to coding costs), or the decisions to be based on the coded data are important.

In this section we emphasize the control function—identifying coding problems early and correcting them before they lead to a broad base of complications. The iterative process (code and evaluate, then take corrective actions and reevaluate) is most helpful if employed at several points early in the judgment/coding process, rather than reserved until the end when correction of problems means starting the whole coding process over from scratch.

If a multijudge evaluation on a subsample of responses shows that the reliability of the coding process is high (e.g., $I_r > .90$), it may be reasonable to complete the coding process with only one judge per response. However, one must keep in mind that the reliability is an upper bound on the validity and thus the single-judge approach is potentially dangerous. Even with relatively high reliability estimates based on a test sample, some of the data included in the final analysis will be in error when only one judge is used in evaluating stimuli. The use of multiple judges—accompanied by reliability indices computed for diagnostic and evaluative purposes—does not guarantee that the data will be valid and reliable. The researcher must be concerned with the quality of the raw responses as well as with the coding. However, the use of multiple judges and evaluation procedures like those recommended here probably affords the best available safeguards against the hidden, and sometime insidious, problems of poorly coded data.

REFERENCES

Armstrong, Gary M. (1984), "An Evaluation of the Children's Advertising Review Unit," *Journal of Public Policy and Marketing*, 3, 38–55.

Assmus, Gert, John U. Farley, and Donald R. Lehmann (1984), "How Advertising Affects Sales: Meta-Analysis of Econometric Results," *Journal of Marketing Research*, 21 (February), 65–74.

Bennett, E. M., R. Alpert, and A. C. Goldstein (1954), "Communications Through Limited Response Questioning," *Public Opinion Quarterly*, 18, 303–8.

Bettman, James R. and C. Whan Park (1980), "Effects of Prior Knowledge and Experience and Phase of the Choice Process on Consumer Decision Processes: A Protocol Analysis," *Journal of Consumer Research*, 7 (December), 234–48.

——— and Barton A. Weitz (1983), "Attributions in the Board Room: Causal Reasoning in Corporate Annual Reports," *Administrative Science Quarterly*, 28 (June), 165–83.

Biehal, Gabriel and Dipankar Chakravarti (1982), "Experiences with the Bettman and Park Verbal-Protocol Coding Scheme," *Journal of Consumer Research*, 8 (March), 442–48.

Brennan, Robert L. and Dale J. Prediger (1981), "Coefficient Kappa: Some Uses, Misuses, and Alternatives, *Educational and Psychological Measurement*, 41, 687–99.

Brucks, Merrie, Gary M. Armstrong, and Marvin Goldberg (1988), "Children's Use of Cognitive Defenses Against Television Advertising," *Journal of Consumer Research*, 14 (March), 471–82.

Churchill, Gilbert A., Jr. (1979), "A Paradigm for Developing Better Measures of Marketing Constructs," *Journal of Marketing Research*, 16 (February), 64–73.

———, Neil M. Ford, Steven W. Hartley, and Orville C. Walker, Jr. (1985), "The Determinants of Salesperson Performance: A Meta-Analysis," *Journal of Marketing Research*, 22 (May), 103–18.

Cohen, Jacob (1960), "A Coefficient of Agreement for Nominal Scales," *Educational and Psychological Measurement*, 20, 37–46.

——— (1968), "Weighted Kappa: Nominal Scale Agreement with Provision for Scaled Disagreement or Partial Credit, *Psychological Bulletin,* 70 (October), 213–20.

Crittenden, K. S. and R. J. Hill (1971), "Coding Reliability and Validity of Interview Data," *American Sociological Review,* 36, 1073–80.

Dewey, M. E. (1983), "Coefficients of Agreement," *British Journal of Psychiatry,* 143, 487–9.

Didow, Nicholas M., William D. Perreault, Jr., and Nicholas Williamson (1983), "A Cross-Sectional Optimal Scaling Analysis of the Index of Consumer Sentiment," *Journal of Consumer Research,* 10 (December), 339–47.

Dillon, William R. and Narendra Mulani (1984), "A Probabilistic Latent Class Model for Assessing Inter-Judge Reliability," *Multivariate Behavioral Research,* 19 (October), 438–58.

Everitt, B. S. (1968), "Moments of the Statistics Kappa and Weighted Kappa," *British Journal of Mathematical and Statistical Psychology,* 21 (May), 97–103.

Farley, John U. and R. L. Swinth (1967), "Effects of Choice and Sales Messages on Customer-Salesman Interaction," *Journal of Applied Psychology,* 52 (April), 107–10.

Fern, Edward F., Kent B. Monroe, and Ramon A. Avila (1986), "Effectiveness of Multiple Request Strategies: A Synthesis of Research Results," *Journal of Marketing Research,* 23 (May), 144–52.

Fleiss, Joseph L. (1971), "Measuring Nominal Scale Agreement Among Many Raters," *Psychological Bulletin,* 76 (5), 378–82.

——— (1975), "Measuring Agreement Between Two Judges on the Presence or Absence of a Trait," *Biometrics,* 31, 651–9.

———, Jacob Cohen, and B. S. Everitt (1969), "Large Sample Standard Errors of Kappa and Weighted Kappa," *Psychological Bulletin,* 72 (5), 323–7.

Franke, George R. and Nicholas M. Didow (1987), "Phenomenological Validity in Marketing Research: Self Coding Versus Independent Judge Coding of Ambiguous Response Measures," abstracted in AMA Educators' Conference Proceedings. Chicago: American Marketing Association, p. 237.

Goodman, Leo A. and William H. Kruskal (1954), "Measures of Association for Cross Classifications," *American Statistical Association Journal,* 49, 732–64.

Green, Paul E., Frank J. Carmone, and David P. Wachspress (1977), "On the Analysis of Qualitative Data in Marketing Research," *Journal of Marketing Research,* 14 (February), 52–9.

Helzer, John E., Lee N. Robins, Mitchell Tarbleson, Robert A. Woodruff, Theodore Reich, and Eric D. Wish (1977), "Reliability of Psychiatric Diagnosis: A Methodological Review," *Archives of General Psychiatry,* 34 (February), 129–33.

Hoffman, Donna L. and George R. Franke (1986), "Correspondence Analysis: Graphical Representation of Categorical Data in Marketing Research," *Journal of Marketing Research,* 23 (August), 212–27.

Holsti, O. (1969), *Content Analysis for the Social Sciences and the Humanities*. Reading, MA: Addison-Wesley Publishing Company.

Hubert, Lawrence (1977), "Kappa Revisited," *Psychological Bulletin,* 84 (March), 289–97.

Janes, Cynthia L. (1979), "An Extension of the Random Error Coefficient of Agreement to N * N Tables," *British Journal of Psychiatry,* 134, 617–19.

Koran, Lorrin M. (1975), "The Reliability of Clinical Methods, Data, and Judgments," *New England Journal of Medicine,* 293 (October 2), 695–701.

Kraemer, Helena C. (1979), "Ramifications of a Population Model for K as a Coefficient of Agreement," *Psychometrica,* 44 (December), 461–72.

Krippendorff, K. (1970), "Bivariate Agreement Coefficients for Reliability of Data," in *Sociological Methodology,* E. F. Borgatta and G. W. Bohrnstedt, eds. San Francisco: Jossey-Bass Publishers.

Lammers, H. Bruce (1985), "A Multitrait-Multimethod Analysis of the Validity of Cognitive Response Assessment Procedures," in *Advances in Consumer Research,* Vol. 12, E. Hirschman and M. Holbrook, eds. Ann Arbor, MI: Association for Consumer Research, 164–8.

Landis, J. Richard and Gary G. Koch (1977), "The Measurement of Observer Agreement for Categorical Data," *Biometrics,* 33 (March), 159–74.

Lawlis, G. G. and E. Lu (1972), "Judgments of Counseling Process: Reliability, Agreement, and Error," *Psychological Bulletin,* 78 (1), 17–20.

Light, Richard J. (1971), "Measures of Response Agreement for Qualitative Data: Some Generalizations and Alternatives," *Psychological Bulletin,* 76 (5), 365–77.

Maxwell, A. E. (1977), "Coefficients of Agreement Between Observers and Their Interpretation," *British Journal of Psychiatry,* 130, 79–83.

Mitchell, Sandra K. (1979), "Interobserver Agreement, Reliability, and Generalizability of Data Collected in Observational Studies," *Psychological Bulletin,* 86 (2), 376–90.

Perreault, William D., Jr. and Forrest W. Young (1980), "Alternating Least Squares Optimal Scaling: Analysis of Nonmetric Data in Marketing Research," *Journal of Marketing Research,* 17 (February), 1–13.

——— and Hiram C. Barksdale, Jr. (1980), "A Model-Free Approach for Analysis of Complex Contingency Data in Survey Research," *Journal of Marketing Research,* 17 (November), 503–15.

Peter, J. Paul (1979), "Reliability: A Review of Psychometric Basics and Recent Marketing Practices," *Journal of Marketing Research,* 16 (February), 6–17.

——— (1981), "Construct Validity: A Review of Basic Issues and Marketing Practices," *Journal of Marketing Research,* 18 (May), 133–45.

——— and Gilbert A. Churchill, Jr. (1986), "Relationships Among Research Design Choices and Psychometric Properties of Rating Scales: A Meta-Analysis," *Journal of Marketing Research,* 23 (February), 1–10.

——— and Michael L. Ray (1984), *Measurement Readings for Marketing Research*. Chicago: American Marketing Association.

Petty, Richard E., Thomas M. Ostrom, and Timothy C. Brock (1981), *Cognitive Responses in Persuasion*. Hillsdale, NJ: Lawrence Erlbaum Associates.

Scott, W. A. (1955), "Reliability of Content Analysis: The Case of Nominal Scale Coding," *Public Opinion Quarterly,* 19 (Fall), 321–25.

Spiegel, Murray R. (1961), *Theory and Problems of Statistics*. New York: McGraw-Hill Book Company.

Wright, Peter (1973), "The Cognitive Processes Mediating Acceptance of Advertising," *Journal of Marketing Research,* 10 (February), 53–62.

Reprint No. JMR262100

Copyright of Journal of Marketing Research (JMR) is the property of American Marketing Association and its content may not be copied or emailed to multiple sites or posted to a listserv without the copyright holder's express written permission. However, users may print, download, or email articles for individual use.